The Tug-of-War Mystery!

A Story About Balanced and Unbalanced Forces

The grand parade at Wonder Pier was over, but Maya and Ethan weren't ready to leave.

Suddenly, they heard loud cheering near the carnival's sports arena.

A tug-of-war competition was taking place!

Two teams were pulling a long rope in opposite directions. Surprisingly, the rope wasn't moving.

"Why isn't either team winning?" Ethan wondered.

"Because both teams are pulling with equal force in opposite directions!" Mr. Oliver explained.

"So the forces cancel each other?" Maya asked.

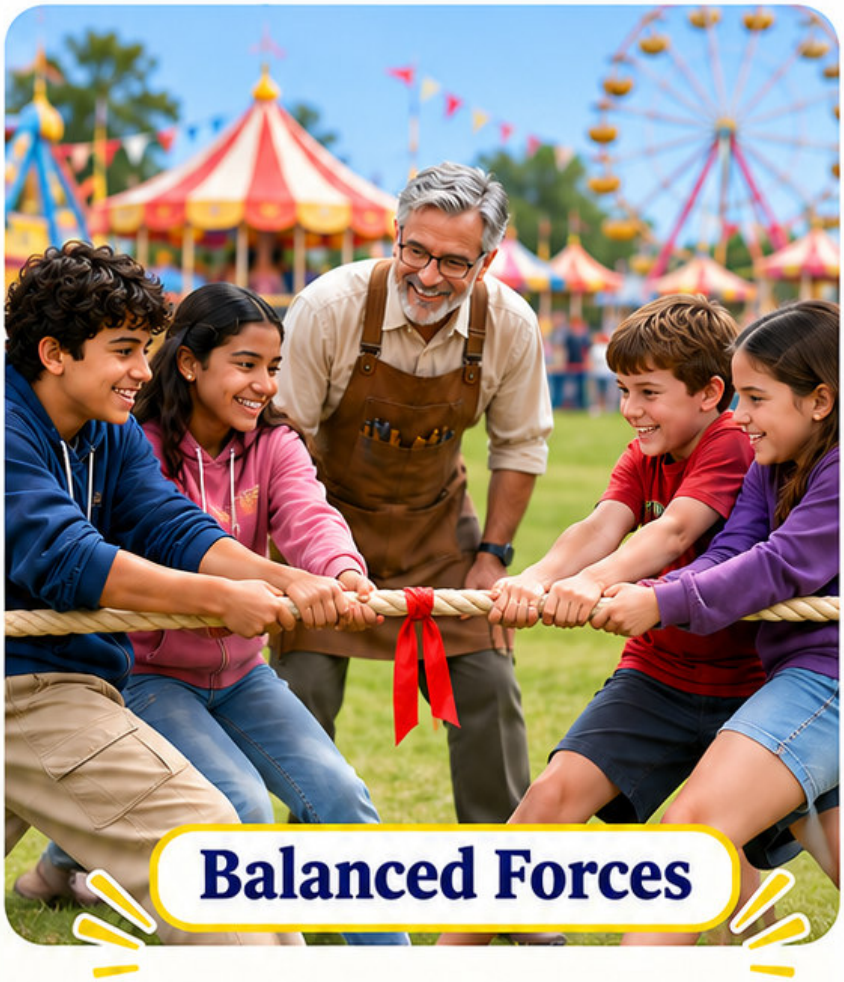
"Exactly! These are called **balanced forces**. When forces are balanced, the net force is zero."

Ethan watched the rope carefully.

"So balanced forces keep a stationary object from starting to move!"

Mr. Oliver smiled. "That's right! But balanced forces can also act on moving objects."

The friends were curious to discover how!



Equal forces in opposite directions



Net force = 0 N (balanced)

Science Fact: Balanced forces cancel each other, producing zero net force. They do not change an object's state of motion.

Science Fables

The Cart That Kept Rolling!

Mr. Oliver led the friends toward a small parade cart on a straight path.

"Watch closely," he said, giving the cart a stronger push.

The cart started moving.

"My push made it start moving because, at first, the forces were unbalanced," Mr. Oliver explained.

Then he gently kept pushing, and the cart rolled forward at a steady speed.

"Why isn't it speeding up?" Maya asked.

"Because now my push balances the friction and other forces slowing it down," Mr. Oliver explained.

Ethan looked surprised. "So the cart can move even when the forces are balanced?"

"Exactly! Balanced forces don't always mean standing still. They mean no change in motion. A moving object can keep going at a constant speed in a straight line."

Maya smiled. "So an unbalanced force started the cart, and balanced forces kept it moving steadily!"

"Correct!" Mr. Oliver replied.





Science Fact: An unbalanced force can start motion. Balanced forces can then keep a moving object at constant velocity.

Science Fables

When One Team Pulls Harder!

The friends returned to the tug-of-war arena for the final round.

This time, something exciting happened!

One team pulled much harder than the other, and the rope suddenly moved toward them.

“Look! The rope is moving!” Ethan shouted.

“That’s because the forces are no longer balanced,” Mr. Oliver explained.

“When forces do not cancel each other, they’re called unbalanced forces. The net force is not zero.”

“So the rope moves toward the team applying the greater force?” Maya asked.

“Exactly! Unbalanced forces can change an object’s speed or direction.”

Ethan grinned. “Now I understand why one team wins!”



Unbalanced Forces



Net force = 30 N
(to the left)

Science Fact: Unbalanced forces produce a nonzero net force and cause a change in an object’s motion.

Science Fables

The Mystery of the Slowing Cart!

Next, Mr. Oliver challenged the friends to another experiment.

He placed a toy car on ground and gave it a push.

The car began rolling forward.

“The push made it start moving!” Maya exclaimed.

But after a short distance, the car slowed down and stopped.

“Why didn’t it keep moving?” Ethan wondered.

“Because friction acts against its movement,” Mr. Oliver explained. “Once I stopped pushing, friction created an unbalanced force that slowed the car down.”

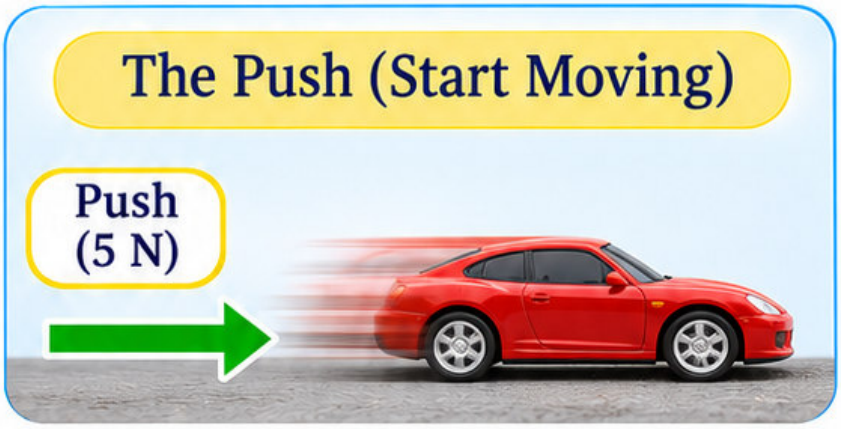
Maya thought for a moment.

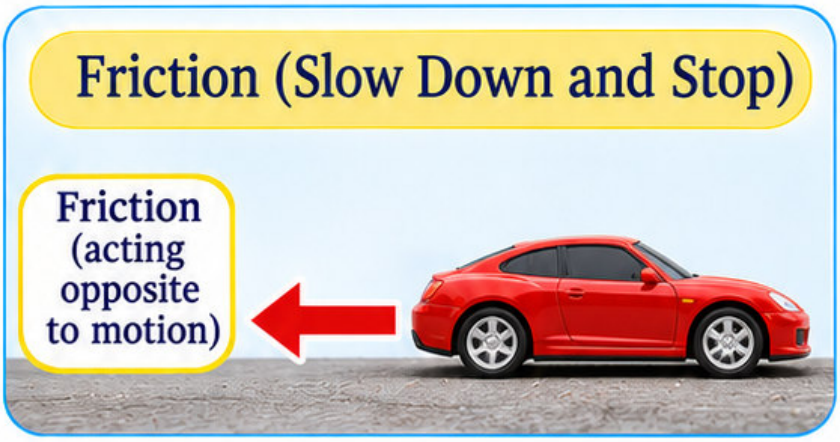
“So unbalanced forces can make an object start moving, speed up, or slow down!”

“And they can change its direction too!” Mr. Oliver added.

Ethan smiled. “Unbalanced forces really change how things move!”







Science Fact: Unbalanced forces can start or stop motion, change speed, or change direction.

Science Fables

The Final Force Challenge!

Before leaving Wonder Pier, Mr. Oliver showed the friends two force arrows on a screen.

One arrow pointed right with 5 N, while the other pointed left with 5 N.

“Are these forces balanced?” he asked.

“Yes!” Maya answered. “They’re equal and opposite, so the net force is zero!”

Mr. Oliver changed the arrows.

Now one showed 7 N to the right, while the other showed 4 N to the left.

“What about now?” he asked.

Ethan quickly answered, “They’re unbalanced! The net force is 3 N to the right!”

“Perfect!” Mr. Oliver exclaimed. “The net force tells us whether an object’s motion will change.”

Maya smiled. “Now we know that forces can work together or oppose each other!”

As the carnival lights began to glow, the friends waved goodbye.

Their tug-of-war mystery was finally solved!



Balanced Forces



Net force = 0 N (balanced)

Unbalanced Forces



Net force = 3 N (to the right)

Science Fact: Net force is the overall force acting on an object. Balanced forces have zero net force, while unbalanced forces have a nonzero net force.

Science Fables

Science Element: Remember!

1. What Are Balanced and Unbalanced Forces?

Balanced Forces	Unbalanced Forces
Forces cancel each other	Forces do not cancel
Net force = 0 N	Net force ≠ 0 N
No change in velocity	Velocity changes
Object remains at rest or moves with constant velocity	Object can start, stop, speed up, slow down, or change direction

2. Understanding Net Force

Net force is the overall force acting on an object.
When forces act in opposite directions, subtract the smaller force from the larger force.

Forces	Net Force	Result
5 N right, 5 N left	0 N	Balanced
7 N right, 4 N left	3 N right	Unbalanced
8 N left, 3 N right	5 N left	Unbalanced

3. Everyday Examples

Balanced Forces	Unbalanced Forces
Equal teams pulling a stationary rope	One tug-of-war team pulling harder
A book resting on a table	Kicking a stationary ball
A car traveling at constant speed on a straight road	A rolling ball slowing because of friction

4. Effects of Unbalanced Forces

An unbalanced force can:

- Start or stop motion.
- Increase or decrease speed.
- Change direction.



5. The Golden Rule

Remember!

- Balanced Forces: Net force = 0 N. No change in velocity.
- Unbalanced Forces: Net force ≠ 0 N. Velocity changes.
- Equal and opposite forces: Cancel each other.
- Unequal opposite forces: Produce a net force toward the stronger force.
- Important: An object can still be moving when the forces acting on it are balanced!

Science Fables